1. ECDLP lead pursuit audit

160-lead reconciled atlas · 2026-11-01



1.1. Decision

The 159-row ECDLP atlas at `crypto-autoresearcher@7072741f712c` was reviewed family by family against `crypto@b6ffdca53494`. The review also added one missing live split-graph/secant successor, so the reconciled atlas has **160 leads**. Exactly **21 narrowly scoped mechanisms** have mechanism-wide obstruction evidence under their stated assumptions. The other **139 leads remain pursued** through a next action or explicit revisit trigger.

This is a scope decision. Each stopped row is stopped only for the exact mechanism in its obstruction. A different presentation, locus, oracle, statistic, or stated escape remains live.

1.2. Queue invariant

Only `closed_obstructed` leaves the pursuit queue. `closed_known`, `closed_dominated`, and `exhausted_at_scale` remain queued through their revisit triggers. Timeouts, OOMs, missing infrastructure, unavailable sources, finite nulls, and finite measured cost disadvantages are not mechanism-wide disproof. A complete lower bound inside an explicitly named model obstructs only that model.

When a row combined a proved negative subcase with an untested escape, the audit preserved the obstruction as evidence and restored the row to `open_next_step` using its existing `revisit_when`. Twenty-six overbroad closures were corrected this way.

1.3. Family accounting

Family	Reviewed	Obstructed	Pursued
Binary-field IC	23	2	21
IC engines	20	2	18
Prime-field IC	27	3	24
Generic, transfer, structure, exotic	90	14	76
Total	**160**	**21**	**139**

1.4. Exact suppressed scopes

- Binary IC: `generic-table-oracle`; linear/subspace scope of `frobenius-orbit-factor-base`.
- IC engines: coordinate-blind scope of `generic-oracle-arity-cost-floor`; generic-fibre census of `semaev-cover-monodromy`.
- Prime IC: `generic-table-decomposition`; generic equality-filter scope of `representation-subset-sum`; `self-pairing-prime-order`.
- Generic: `generic-sqrt-lower-bound`; generic auxiliary-power acquisition in `cheon-aux-acquisition`.
- Transfer: the twelve fixed-arity mechanisms in `cm-canonical-lift-heights`; `one-point-self-pairing`; homomorphic scope of `padic-canonical-section`; class-invariant scope of `prime-field-isogeny-to-weak`; homomorphic scope of `second-torsion-direction`.

- Structure: generic scopes of `endo-scalar-action-generic`, `isogeny-class-generic-invariance`, and `isogeny-transport-preprocessing`; class-invariant weaknesses in `isogeny-walk-weak-neighbour`; scalar/generic scopes of `p192-class-cm-relations` and `secp256k1-cm-class-audit`.
- Exotic: the named GGM-simulable scope of `augmented-oracle-ggm`.

1.5. Load-bearing corrections

The binary SAT/enumeration measurements do not test symmetric Groebner systems at `ell=6..8`, high-arity closure, or the GG14 `m=4` presentation. The Semaev chain identity assumes rather than measures topology-independent closure degree. Finite volcano, endomorphism, factor-base, and residual-relation panels do not prove absence beyond their tested regimes. Proofs for homomorphic, bounded-degree, generically Galois, or full-Jacobian routes do not close explicitly non-homomorphic, growing-degree, non-Galois, or projected successors.

Load-bearing cross-checks are `crypto:research/factor_base_arity_review_20261024/README.md`, `crypto:research/prime_factor_base_degree_20261031/parallel_20261031/RESULTS.md`, `crypto:problems/ghs-higher-magic-descent-not-executable.md`, and every atlas row's obstruction, foreclosure, revisit, evidence, and related fields.

No performance result is introduced. This report records disposition and queue semantics only.

1.6. Xedni correction

The later primary-source and kernel-checked review at `crypto@34caae22581045ec59fe779d86f8bf2d495cc14a` found that two xedni rows exceeded JKSST's theorem. Theorem 4.1 says "under certain plausible assumptions"; Lemma 4.1 assumes the selected-lift height/discriminant comparison and Lang, while reduction, useful-relation, cofactor, and adaptive-hazard bridges remain explicit premises.

`LEAD-TRF-global-lift-small-height` and `LEAD-ICP-char0-lifting-xedni-smoothness` are therefore `open_next_step`. Their exact and finite negatives remain recorded. `KR-IC-cd159a` now states the conditional theorem, and `LEAD-EXO-global-lift-height` pursues both the function-field timing and characteristic-zero bridge-composition branches.

The CM row stays suppressed only inside corrected L9: ordinary prime-field Serre–Tate/global CM models, fixed arity 2..9, absolute `C0`, and twelve enumerated mechanisms. Target-dependent arbitrary sections, growing arity or `C0`, and unenumerated families remain outside. The 21/139 totals are unchanged because `closed_known` rows were already pursued.

Primary JKSST PDF SHA-256: `981bf955f7aecc5df4f271837f6653bd7666e5026f59df44196609d4a1dd6bc7`; checked at Section 4, Lemma 4.1, Theorem 4.1, Sections 5–6, and notes 7–8. The sibling proof map is `research/jksst_xedni_audit_20261101/PROOF_MAP.md`.

1.7. Review boundary

The Conductor command was unavailable and no `.conductor/WORKFLOW.md` was present. The global review stamp remains unchanged: the cryptanalysis base ref was unreachable and the configured `ecc-challenges` checkout was absent, so a whole-universe freshness claim would be unsupported. The 159 pre-existing rows, the added live successor, and the cited crypto snapshot are reconciled; the unavailable repositories must be reconciled before advancing the canonical stamp.